Cybersecurity - Graduate Level Assessment 14 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. ____________________________ began to show up few years back on wireless access points as a new way of adding or connecting new devices.
   (a) WPA 2
   (b) WPA
   (c) WPS
   (d) WEP
2. How do you prevent SQL injection?
   (a) Escape queries
   (b) Interrupt requests
   (c) Merge tables
   (d) All of the above
3. Which among them has the strongest wireless security?
   (a) WEP
   (b) WPA
   (c) WPA 2
   (d) WPA 3
4. How does a buffer overflow on the stack facilitate running attacker-injected code?
   (a) By overwriting the return address to point to the location of that code
   (b) By writing directly to the instruction pointer register the address of the code
   (c) By writing directly to %eax the address of the code
   (d) By changing the name of the running executable, stored on the stack
5. Man in the middle attack can endanger the security of Diffie Hellman method if two parties are not
   (a) Joined
   (b) Authenticated
   (c) Submitted
   (d) Shared

True / False (10 marks)

Answer True or False

6. True or False: Brute-force attacks on access credentials typically occur at the data-link layer of the OSI model due to network traffic interception.

7. True or False: When the STP solver times out on a constraint query, EXE assumes the query is not satisfiable and stops executing the path.

8. True or False: The mishandling of poorly defined variables is a recognized transport layer vulnerability.

9. True or False: WPA primarily relies on AES encryption for securing wireless communications.

10. True or False: A MAC with a fixed output length of 5 bits is insecure, as an attacker can easily guess the tag for any message.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the vulnerabilities of SQL queries based on user input to injection attacks, providing examples and analyzing potential mitigation strategies in cybersecurity.

12. Discuss the various operating modes of block ciphers in cybersecurity, highlighting their functions and applications. Identify and analyze a mode that is not commonly recognized as valid.

End of Paper